

Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

```
clc;
```

```
clear;
```

```
close all;
```

```
%% --- Input Parameters ---
```

```
fp = 1000;      % High-pass cutoff frequency (Hz)
```

```
Fs = 8000;      % Sampling frequency (Hz)
```

```
N = 50;         % Filter order
```

```
% Normalized cutoff frequency
```

```
wc = fp/(Fs/2);
```

```
%% --- FIR High-Pass Filter using Hanning Window ---
```

```
window = hann(N+1);
```

```
b_hann = fir1(N, wc, 'high', window);
```

```
%% --- Display Filter Coefficients ---
```

```
disp('--- High-Pass FIR Coefficients using Hanning Window ---');
```

```
disp(b_hann');
```

```
%% --- Frequency Response ---
```

```
figure;
```

```
freqz(b_hann, 1, 1024, Fs);
```

```
title('FIR High-Pass Filter using Hanning Window');
```

```
grid on;
```

```
%% --- Magnitude Response ---
```

```
figure;
```

```
[H, f] = freqz(b_hann, 1, 1024, Fs);
```

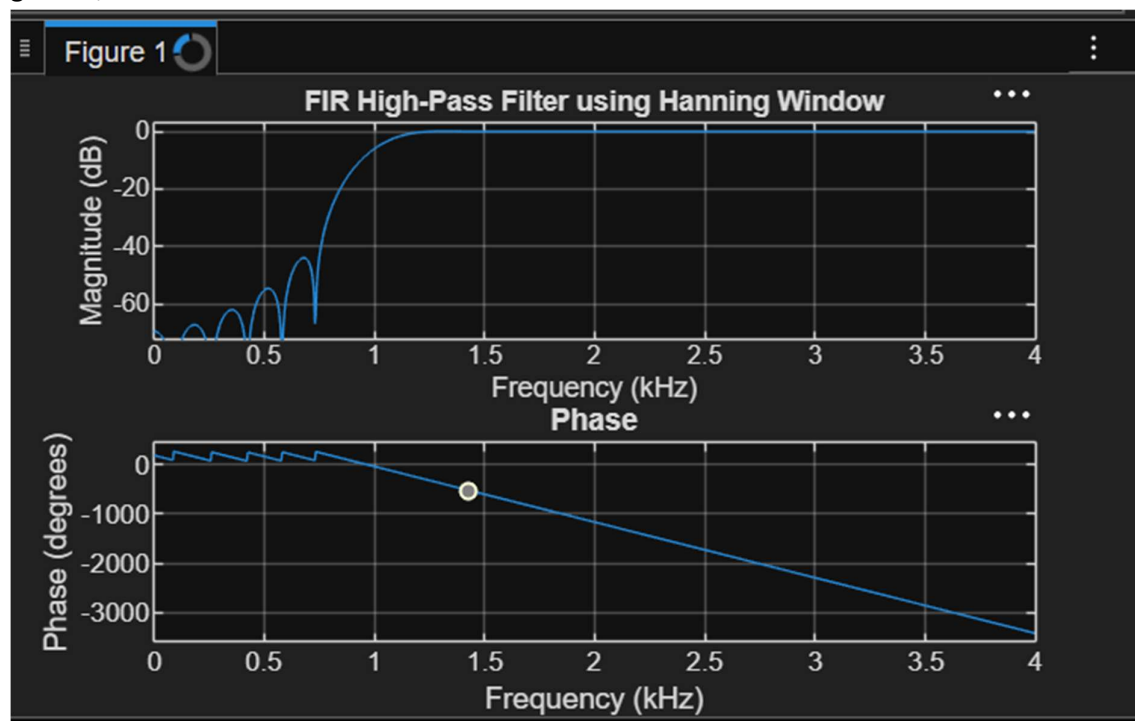
```
plot(f, abs(H), 'LineWidth', 1.5);
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude');
```

```
title('Magnitude Response of FIR High-Pass Filter');
```

```
grid on;
```



Result 1. The FIR High-Pass Filter was successfully designed using the Hanning Window technique for the specified cutoff frequency, sampling frequency, and filter order.

2. The FIR filter coefficients were successfully generated using MATLAB and the designed filter was implemented.

3. The magnitude response and phase response of the designed FIR High-Pass Filter were obtained and analyzed. The magnitude response exhibited attenuation of low-frequency components and passage of high-frequency components, confirming the desired high-pass